ST MARY'S DSG, KLOOF

GRADE: 10 18 February 2021

MATHEMATICS TEST
Products, Factors, Fractions

TIME: 40 min TOTAL: 37 MARKS

Name: Memo

QUESTION 1:

Expand the following and simplify fully:

a) $(x+6)(x^2-6x+36)$ (2)

$= x^3 + 216$

b) $-(x-2)^2$ (3)

$= -(x^2 - 4x + 4)$

$= -x^2 + 4x - 4$

c) $\left(3x-\frac{1}{x}\right)\left(3x+\frac{1}{x}\right)$ (3)

$= 9x^2 - \frac{1}{x^2}$

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QUESTION 2:

Factorise completely:

a) $x^2 - 7x - 18$ (2)

$= (x + 2)(x - 9)$

b) $27y^3 - 8$ (3)

$= (3y - 2)(9y^2 + 6y + 4)$

c) $x^4 - y^4$ (3)

$= (x^2 + y^2)(x^2 - y^2)$

$= (x^2 + y^2)(x - y)(x + y)$

d) $(x + y)^2 + 8(x + y) + 15$ (3)

let $x + y = k$

$= (x + y + 5)(x + y + 3)$

$k^2 + 8k + 15$

$= (k + 5)(k + 3)$

e) $b^2 - c^2 - b - c$ (4)

$= (b - c)(b + c) - (b + c)$

$= (b + c)(b - c - 1)$

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QUESTION 3:

Simplify the following fractions:

a) $\frac{x^2+2x}{x^2-x-6} \times \frac{x-3}{x}$ (3)

$= \frac{x(x+2)}{(x-3)(x+2)} \times \frac{x-3}{x}$

$= 1$

b) $\frac{9-3a}{a^2-4} \div \frac{3a^2-11a+6}{a+2}$ (5)

$= \frac{3(3-a)}{(a-2)(a+2)} \times \frac{a+2}{(3a-2)(a-3)}$

$= \frac{-3}{(a-2)(3a-2)}$

[8]

QUESTION 4:

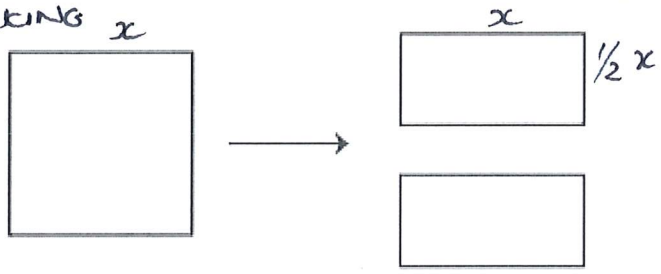
a) If $x^2 + 2x - 5 = (x + 1)^2 + k$, find the value of k. (2)

$x^2 + 2x - 5 = x^2 + 2x + 1 + k$

$\therefore k = -6$

b) A square is cut into two identical rectangles, as shown. Each of these two rectangles has perimeter 24 cm. What is the area of the original square? (4)

SHOW WORKING



$2(\frac{1}{2}x) + x + x = 3x$

$3x = 24$

$x = 8$

$\therefore$ Area $= 8^2 = 64\ cm^2$

[6]